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Cryptography | Technical Presentation

Steganography vs Cryptography

Hidden Security in the Digital World

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Two ways to protect a message

Cryptography

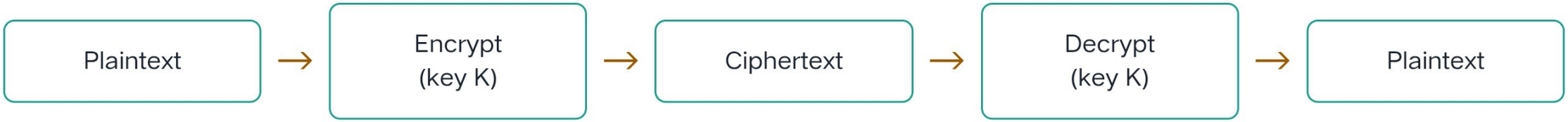
- Makes the content unreadable
- Needs a key to recover the message
- Attacker knows a secret exists
- Strong, well-studied algorithms (AES, RSA)
- Protects confidentiality, integrity, authenticity

Steganography

- Hides the message in an ordinary file
- Attacker should see nothing unusual
- Works on images, audio, text and video
- Capacity is limited by the cover
- Protects the existence of the message

Together they answer both questions an attacker asks: what does the message say, and is there a message at all?

How cryptography works



Symmetric

- One shared secret key
- Fast: AES, 3DES, ChaCha20
- Problem: key distribution
- Bulk data encryption

Asymmetric

- Public and private key pair
- RSA, ECC, Diffie-Hellman
- Slower, but solves key exchange
- Signatures and key transport

Hash functions

- SHA-256, SHA-3
- One-way, fixed-length digest
- HMAC adds a secret key
- Detects any change in data

Kerckhoffs's principle (1883): security must rest on the key, not on a secret algorithm.

How steganography works



Cover media

- Image (BMP, PNG, JPEG)
- Audio (WAV, MP3)
- Text and documents
- Video and network traffic

Requirements

- Imperceptibility: no visible change
- Capacity: how much can be hidden
- Robustness: survives edits
- Security: resists steganalysis

Prisoners' problem

- Simmons, 1983
- Alice and Bob are the prisoners
- The warden reads every message
- Warden must not get suspicious

From Greek *steganos* (covered) and *graphein* (writing).

Common steganographic techniques

Medium	Method	Weakness
Image	LSB replacement in pixel bytes	Lost on lossy compression
Audio	LSB, echo hiding, phase coding	MP3 re-encoding destroys bits
Text	Whitespace, synonym choice	Very low capacity
Video	Frames, motion vectors	Heavy to process
Network	Unused header fields, timing	Traffic analysis can spot it

Spatial domain

- Change pixel values directly
- Simple, high capacity

Transform domain

- Change DCT or DWT coefficients
- Survives JPEG compression better

Adaptive

- Hide in edges and noisy areas
- Harder to detect

Worked example: LSB embedding of the letter H

H = 01001000 in ASCII. One secret bit replaces the last bit of each pixel byte.

Pixel byte	Binary	Secret bit	New byte
200	11001000	0	200
143	10001111	1	143
96	01100000	0	96
77	01001101	0	76
54	00110110	1	55
181	10110101	0	180
22	00010110	0	22
129	10000001	0	128

4 of 8 bytes changed, each by at most 1, which the eye cannot notice.

Cryptography vs steganography

Aspect	Cryptography	Steganography
Primary goal	Secrecy of content	Secrecy of existence
Attacker sees	Obvious encrypted data	A normal-looking file
Output looks like	Random-looking bytes	Unchanged image or audio
If detected	Still protected by key	Protection usually lost
Key	Essential	Optional stego key
Typical attack	Cryptanalysis, brute force	Steganalysis, statistics
Typical tools	AES, RSA, OpenSSL	Steghide, OpenStego
Real-world use	HTTPS, disk encryption, VPN	Watermarking, covert channels
Main weakness	Draws attention	Low capacity, fragile

Attacks and defences

Cryptanalysis

- Ciphertext-only
- Known-plaintext
- Chosen-plaintext
- Chosen-ciphertext
- Brute force on the key

Steganalysis

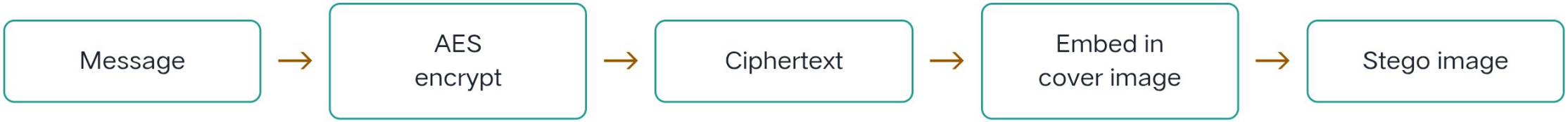
- Visual and histogram inspection
- Chi-square test on LSB pairs
- RS analysis
- File size and metadata checks
- Comparing with the original cover

Countermeasures

- Long random keys (AES-256)
- Encrypt before embedding
- Never reuse a cover image
- Low embedding rate
- Noisy or textured covers

Trade-off: more capacity means less imperceptibility and less robustness.

Using both together



Why combine

- Encrypted bits look random, like noise
- Changed LSBs then look natural
- If hiding is found, the cipher still holds
- Two independent layers of defence

Process

- Sender: encrypt with AES, then embed
- Receiver: extract, then decrypt
- Share the AES key and stego key safely
- Use a lossless cover such as PNG or BMP

Order matters: encrypt first, then embed.

Tools and demonstration plan

Tool	Use in the demo
OpenSSL	AES-256-CBC encryption of the message
Steghide	Embed and extract in JPEG/BMP with a passphrase
Python + Pillow	LSB embedding, pixel-by-pixel comparison
Chi-square script	Detection test on cover vs stego image

Demo steps

- 1. Encrypt the message with OpenSSL
- 2. Embed the ciphertext with Steghide
- 3. Compare cover and stego images
- 4. Run the chi-square detection test
- 5. Extract and decrypt the message

What to show

- Cover and stego image side by side
- Pixel difference of at most 1 per byte
- Histogram before and after embedding
- Detection result for plain vs encrypted payload

Conclusion and references

Conclusion

- Cryptography protects the content
- Steganography protects the existence
- Steganography alone is weak once found
- Strongest design: encrypt first, then hide

Applications

- Secure messaging and storage
- Watermarking and copyright marking
- Covert-channel research
- Digital forensics

References

- Stallings, Cryptography and Network Security
- Kerckhoffs, La cryptographie militaire, 1883
- Simmons, The Prisoners' Problem and the Subliminal Channel, 1983
- Provos and Honeyman, Hide and Seek, IEEE Security & Privacy, 2003